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 # Project Title: Mobile Phone Price Analysis & Prediction Using Data Analytics

Problem Statement : I am analyzing the Mobile Phone Price Classification Dataset to understand which features (like RAM, battery power, screen size, etc.) influence the price range of a smartphone. The goal is to clean the dataset, analyze relationships between features, visualize trends, and finally build a simple model that predicts the price category of a phone. This can help smartphone companies understand what features increase price.

 # Dataset Description:

Dataset Name: Mobile Price Classification Dataset

Source: Kaggle

Rows: 2000; Columns: 21

Dataset Link - https://www.kaggle.com/datasets/navjotkaushal/mobile-price-classification-dataset

Final insights:

Finding 1:

- The histogram in Section 3 shows that RAM increases gradually, and higher RAM values correspond to higher price ranges.

Finding 2:

- The heatmap shows that RAM has the strongest correlation with price. This means RAM is the biggest factor affecting mobile price.

Finding 3:

- The boxplot shows outliers in battery power, indicating some phones have unusually high battery capacity.

Final Recommendations:

- Companies should focus on increasing RAM if they want to create higher-priced premium phones.
- Battery capacity also influences price—phones with stronger batteries can be marketed as "long battery life" devices.

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score
from sklearn.linear_model import LogisticRegression

# Load dataset
df = pd.read_excel("Mobile Price Classification Dataset.xlsx")
df.head()
```

	Battery_power_mAh	Bluetooh	Speed_of_microprocessor	Dual_sim	Front_came
0	842 mAh	No	2.2	No	1 pix
1	1021 mAh	Yes	0.5	Yes	0 pix
2	563 mAh	Yes	0.5	Yes	2 pix
3	615 mAh	Yes	2.5	No	0 pix
4	1821 mAh	Yes	1.2	No	13 pix

5 rows × 21 columns

```
# Check Rows & Columns
df.shape
```

```
(2000, 21)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2000 entries, 0 to 1999
Data columns (total 21 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Battery_power_mAh        2000 non-null   int64
 1   Bluetooh                 2000 non-null   object
 2   Speed_of_microprocessor  2000 non-null   float64
 3   Dual_sim                 2000 non-null   object
 4   Front_camera             2000 non-null   object
 5   4G                       2000 non-null   object
 6   Internal_memeory_gb      2000 non-null   object
 7   Mobile_depth             2000 non-null   object
 8   Mobile_weight            2000 non-null   object
 9   Cores_of_processor       2000 non-null   int64
 10  Primary_camera           2000 non-null   object
```

```
 11  px_height                2000 non-null   object
 12  Pixel_width              2000 non-null   object
 13  Ram_mb                   2000 non-null   int64
 14  Screen_height            2000 non-null   object
 15  Screen_weight            2000 non-null   object
 16  talk_time                2000 non-null   int64
 17  3G                       2000 non-null   object
 18  touch_screen             2000 non-null   object
 19  wifi                     2000 non-null   object
 20  price_range              2000 non-null   object
dtypes: float64(1), int64(4), object(16)
memory usage: 328.3+ KB
```

```
# Check Dataset Info
df.describe()
```

	Speed_of_microprocessor	Cores_of_processor	talk_time
count	2000.000000	2000.000000	2000.000000
mean	1.522250	4.520500	11.011000
std	0.816004	2.287837	5.463955
min	0.500000	1.000000	2.000000
25%	0.700000	3.000000	6.000000
50%	1.500000	4.000000	11.000000
75%	2.200000	7.000000	16.000000
max	3.000000	8.000000	20.000000

```
# Data Types
df.dtypes
```

	0
Battery_power_mAh	object
Bluetooh	object
Speed_of_microprocessor	float64
Dual_sim	object
Front_camera	object
4G	object
Internal_memeory_gb	object
Mobile_depth	object
Mobile_weight	object
Cores_of_processor	int64
Primary_camera	object
px_height	object
Pixel_width	object
Ram_mb	object
Screen_height	object
Screen_weight	object
talk_time	int64
3G	object
touch_screen	object
wifi	object
price_range	object

dtype: object

```
# Missing values
df.isnull().sum()
```

	0
Battery_power_mAh	0
Bluetooh	0
Speed_of_microprocessor	0
Dual_sim	0
Front_camera	0
4G	0
Internal_memeory_gb	0
Mobile_depth	0
Mobile_weight	0
Cores_of_processor	0
Primary_camera	0
px_height	0
Pixel_width	0
Ram_mb	0
Screen_height	0
Screen_weight	0
talk_time	0
3G	0
touch_screen	0
wifi	0
price_range	0

dtype: int64

```
# Remove Dulpicates
df.drop_duplicates(inplace=True)
# There is no duplicates values
```

```
# column Ram_mb is NOT numeric.It contains values like:
#"2549 mb"
#"2631 mb"
#"2603 mb"
# So Python cannot calculate mean/median because the column has text + numbe
# Solution : Convert Ram_mb to pure numeric
#Step 1 - Remove the text " mb"
#Step 2 - Convert the column into integer

df['Ram_mb'] = df['Ram_mb'].str.replace("mb", "", regex=False)
```

```
df['Ram_mb'] = df['Ram_mb'].str.replace(" ", "", regex=False)
df['Ram_mb'] = pd.to_numeric(df['Ram_mb'])
```

```
df['Ram_mb'].head()
```

	Ram_mb
0	2549
1	2631
2	2603
3	2769
4	1411

dtype: int64

```
# battery column Battery_power_mAh might also contain something like "1500 m
df['Battery_power_mAh'].head()
```

	Battery_power_mAh
0	842 mAh
1	1021 mAh
2	563 mAh
3	615 mAh
4	1821 mAh

dtype: object

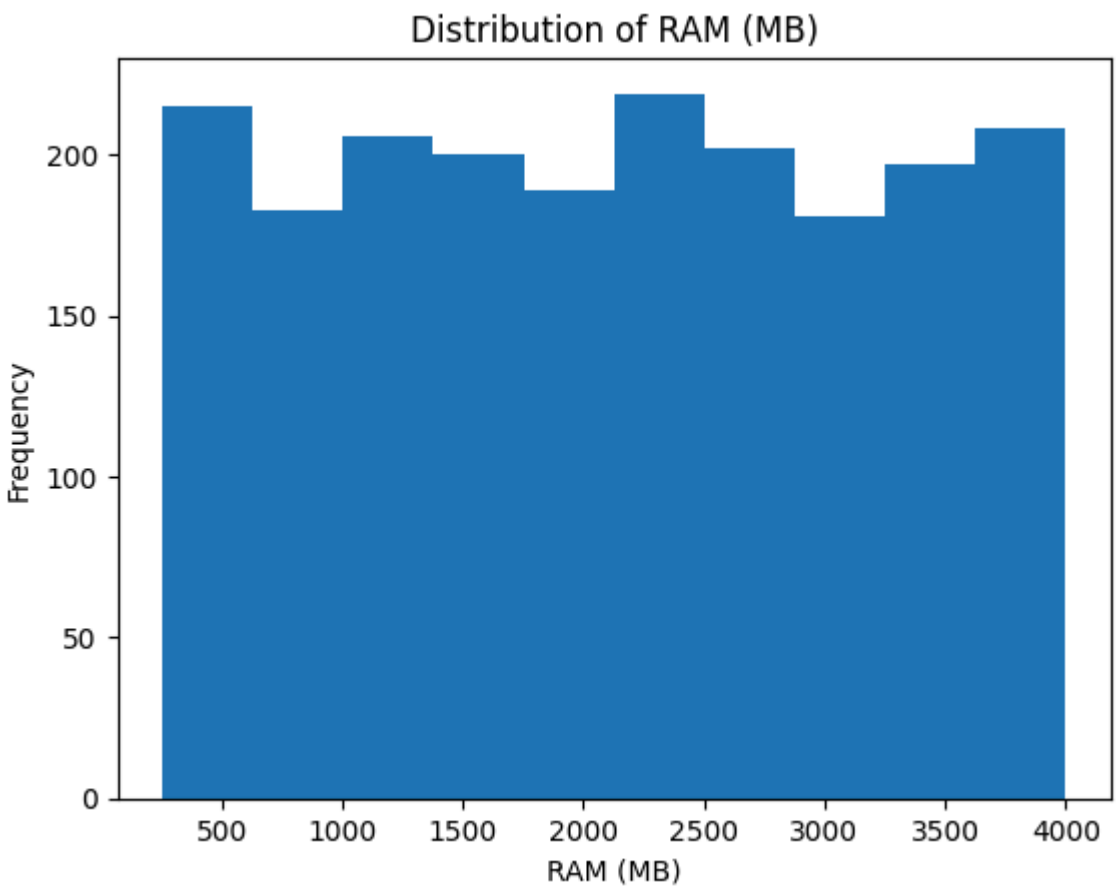
```
# Now run the same for batter also
df['Battery_power_mAh'] = df['Battery_power_mAh'].str.replace("mAh", "", reg
df['Battery_power_mAh'] = df['Battery_power_mAh'].str.replace(" ", "", regex
df['Battery_power_mAh'] = pd.to_numeric(df['Battery_power_mAh'])
```

```
# Remove the text "mAh" & convert the column into integer
df['Battery_power_mAh'].head()
```

	Battery_power_mAh
0	842
1	1021
2	563
3	615
4	1821

dtype: int64

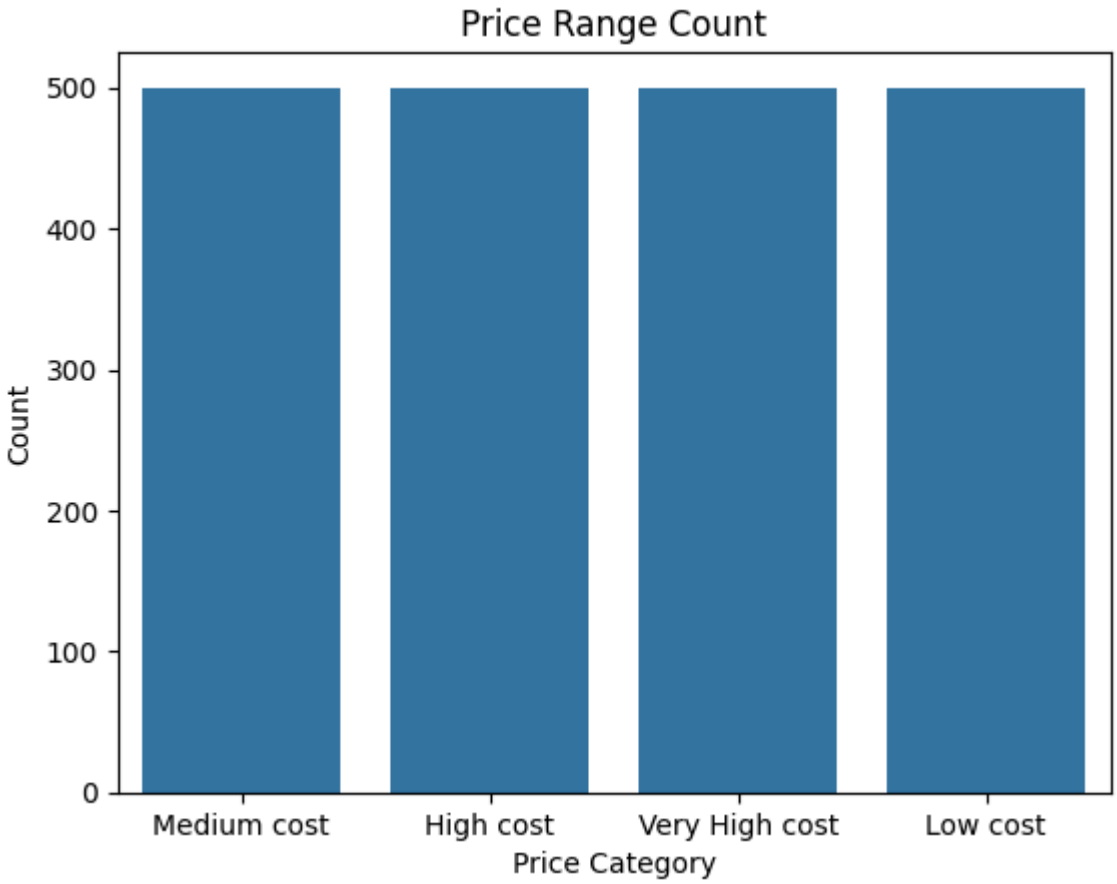
```
# VISUALISATIONS
# Histogram for Numeric (RAM)
import matplotlib.pyplot as plt
plt.hist(df['Ram_mb'])
plt.title("Distribution of RAM (MB)")
plt.xlabel("RAM (MB)")
plt.ylabel("Frequency")
plt.show()

# Explanation
print("The histogram shows how RAM values are distributed, indicating which
```




```
The histogram shows how RAM values are distributed, indicating which RAM size
```

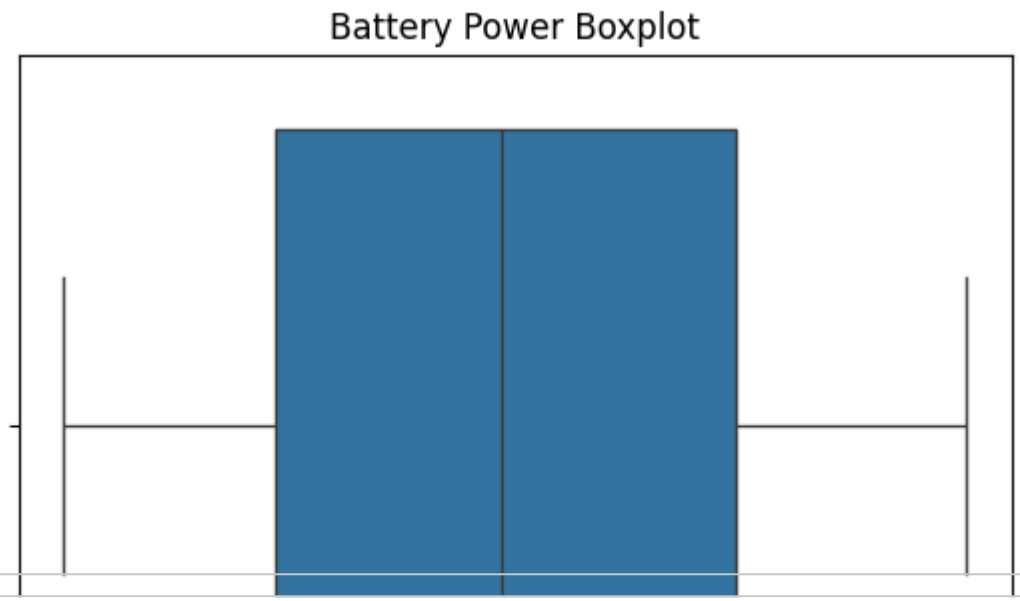
```
# Countplot for Categorical (price_range)
import seaborn as sns
```

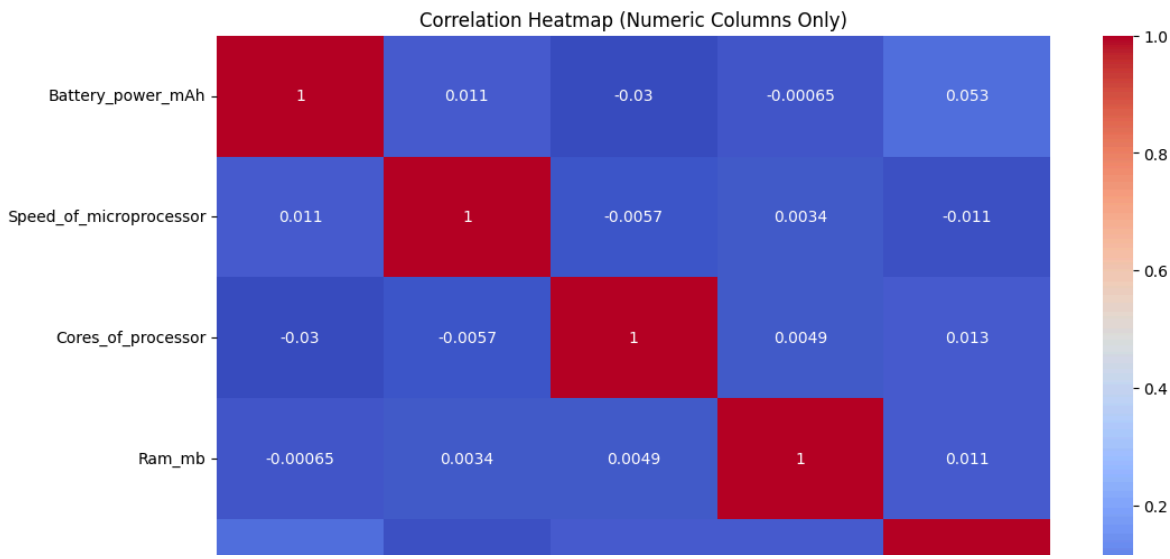
```
sns.countplot(x=df['price_range'])
plt.title("Price Range Count")
plt.xlabel("Price Category")
plt.ylabel("Count")
plt.show()

# Explanation
print("This countplot shows how many mobiles fall into each price range cate
```




```
This countplot shows how many mobiles fall into each price range category.
```

```
# Boxplot for Numeric (Battery Power)
import seaborn as sns
sns.boxplot(x=df['Battery_power_mAh'])
plt.title("Battery Power Boxplot")
plt.xlabel("Battery Power (mAh)")
plt.show()

# Explanation
print("The boxplot highlights the spread of battery power values and shows i
```




```
# Correlation Heatmap (Numeric Only)
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(12,7))

numeric_df = df.select_dtypes(include=['int64', 'float64'])

sns.heatmap(numeric_df.corr(), cmap="coolwarm", annot=True)
plt.title("Correlation Heatmap (Numeric Columns Only)")
plt.show()

# Explanation
print("The heatmap shows how numeric features are related to each other, help
```